

Detailed Syllabus
Lecture-wise Breakup

Subject Code		Semester: (specify Odd/Even):	Semester: 2nd Session: 2025-2026 Month: January – June 2026
Subject Name	RESTful API Design and Implementation		
Credits	3-0-0	Contact Hours	3

Faculty (Names)	Coordinator(s)	
	Teacher(s) (Alphabetically)	

COURSE OUTCOMES		COGNITIVE LEVELS
CO1		
CO2		
CO3		
CO4		
CO5		

Module No.	Subtitle of the Module	Topics in the Module	No. of Lectures for the module
1	Introduction to Web Services & REST Architecture	Evolution of Web Services: SOAP vs REST What is an API? What makes an API RESTful? REST architectural constraints: • Client-Server • Statelessness • Cacheable • Layered System • Uniform Interface • Code-On-Demand (optional) Understanding resources, representations, and actions REST vs GraphQL vs gRPC (brief comparison)	5
2	HTTP Fundamentals for API Development	HTTP protocol overview HTTP methods and their semantics: • GET, POST, PUT, DELETE, PATCH HTTP headers, cookies, and sessions HTTP status codes Request/Response lifecycle Content-Type and Accept headers MIME Types	6
3	Resource Modeling & API Design Principles	Identifying REST resources URI design guidelines Hierarchical vs flat resource structures Using nouns vs verbs in endpoints Designing CRUD operations Query parameters, filters, sorting, and pagination	5

		Hypermedia (HATEOAS) concept API versioning strategies: • URI versioning • Header-based versioning • Query parameter versioning	
4	JSON, Data Formats & Content Negotiation	JSO N structure and schema design XML overview YAML basics Content negotiation using headers Designing standardized request/response bodies Data validation strategies	5
5	API Development – using Java Spring Boot	Setting up the backend project Creating REST controllers/routes Handling request parameters and body JSON serialization/deserialization Service layer implementation Error handling using exceptions/custom error responses Logging and monitoring basics Connecting to databases CRUD APIs with persistent storage File upload/download in REST AP Async API operations	10
6	API Documentation & Testing	Importance of API documentation OpenAPI Specification overview Using Swagger UI/Swagger Edito Generating OpenAPI docs automatically Automated API testing with:Postman, cURL, JUnit/TestNG Mock servers & API simulation Contract testing basics	6
7	Security, Authentication & Authorization	HTTPS & TLS overview Basic Authentication Token-Based Authentication OAuth2 concepts JWT (JSON Web Tokens) – structure & usage Securing endpoints based on roles and permissions CORS: Concepts and configuration API rate limiting & throttling Input sanitization & preventing common attacks: SQL Injection, XSS, CSRF, Authentication bypass Secure API logging and auditing.	5
Total number of Lectures			42
Evaluation Criteria			
Components		Maximum Marks	
T1		20	
T2		20	
End Semester Examination		35	
TA		25 (Attendance (5), Assignment/Tutorial/ Quiz (10), Mini	
Project (10))			
Total		100	

Project based learning: Design and implement RESTful APIs in groups of maximum 4 students each, to illustrate the concepts covered in class.

Recommended Reading material: Author(s), Title, Edition, Publisher, Year of Publication etc. (Text books, Reference Books, Journals, Reports, Websites etc. in the IEEE format)

	Text Books
1	Biehl, M., <i>API Architecture – The Big Picture for Building APIs</i> . API University Press, 2015.
2	Massé, M., <i>REST API Design Rulebook</i> . O'Reilly Media, 2011
	Reference Books
1	Jin, B., Sahni, S., & Shevat, A., <i>Designing Web APIs: Building APIs That Developers Love</i> . O'Reilly Media, 2018.

CO-PO and CO-PSO Mapping:

[illegible]